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NPTEL (https://swayam.gov.in/explorer?ncCode=NPTEL) » The Joy of Computing using Python (course)

Course outline

About NPTEL ()

How does an NPTEL online course work? ()

Week 1 ()

Week 2 ()

Week 3 ()

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Week 5 ()

Week 6 ()

Substitution Cipher -The science of secrecy (unit?unit=124&lesson=125)

Substitution Cipher -The science of

Week 6 : Assignment 6

The due date for submitting this assignment has passed.

Due on 2026-03-04, 23:59 IST.

Assignment submitted on 2026-03-04, 08:23 IST

In-Place Data Reversal in a Memory-Constrained System

Context

A learning analytics platform maintains a list of numerical records representing **student interaction order** during an online examination.The records are stored in the sequence in which events occurred.

For audit, debugging, and review purposes, the system occasionally needs to **reverse the order of these records** so that the **most recent interactions appear first**.

However, the platform operates under **strict memory constraints** because it runs simultaneously for thousands of students.

System Constraints

The development team has imposed the following conditions:

1. The reversal must happen **within the same list** (no new list allowed).

2. The solution must be **function-based**, not written inline.

3. The team wants to explore **non-iterative approaches** as part of internal training.

4. The system should work correctly for:

secrecy 01 (unit?unit=124&lesson=126)
Substitution Cipher -The science of secrecy 02 (unit?unit=124&lesson=127)
Substitution Cipher -The science of secrecy 03 (unit?unit=124&lesson=128)
Tic Tac Toe - Down the memory Lane (unit?unit=124&lesson=129)
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Tic Tac Toe - Down the memory Lane 05 (unit?

o Empty lists

o Lists with odd number of elements

o Lists with even number of elements

Operational Requirement

When a list such as:

[10, 20, 30, 40, 50]

is processed by the system, the **same list object** must be transformed into:

[50, 40, 30, 20, 10]

No additional list should be returned or constructed.

Implementation Provided to the Team

The following implementation was submitted by a developer for review and analysis:

```
def reverseArray(nums):
    start = 0
    end = len(nums) - 1
    swap = 0
    return helper_reverse(start, end, nums, swap)

def helper_reverse(start, end, nums, swap):
    if start > end:
        return nums

    swap = nums[start]
    nums[start] = nums[end]
    nums[end] = swap

    return helper_reverse(start + 1, end - 1, nums, swap)
```

1) If the function reverseArray is called with the input: ***1 point***

nums = [1, 2, 3, 4]

What will be the final value of nums?

- [1, 2, 3, 4]
- [4, 3, 2, 1]
- [2, 1, 4, 3]
- [4, 2, 3, 1]

Yes, the answer is correct.

unit=124&lesson=134)
Recursion (unit?unit=124&lesson=135)
Recursion 01 (unit?unit=124&lesson=136)
Recursion 02 (unit?unit=124&lesson=137)
Recursion 03 (unit?unit=124&lesson=138)
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Quiz: Week 6 : Assignment 6 (assessment?name=654)
Week 6 : Programming Assignment 1 (/noc26_cs84/progassignment?name=626)
Week 6 : Programming Assignment 2 (/noc26_cs84/progassignment?name=627)
Week 6 : Programming Assignment 3

Score: 1
Accepted Answers:
[4, 3, 2, 1]

2) For which condition does the recursive process stop? ***1 point***

- When start == end
- When start > end
- When end == 0
- When swap becomes zero

Yes, the answer is correct.
Score: 1
Accepted Answers:
When start > end

3) What does the function reverseArray(nums) return? ***1 point***

- The modified list nums
- None
- A new reversed list
- Only the last swapped value

Yes, the answer is correct.
Score: 1
Accepted Answers:
The modified list nums

4) What happens when the input list has an odd number of elements? ***1 point***

- The middle element is removed
- The program crashes
- The middle element remains unchanged
- The middle element is swapped with itself twice

Yes, the answer is correct.
Score: 1
Accepted Answers:
The middle element remains unchanged

5) What is the output when reverseArray([]) is executed? ***1 point***

- Runtime error
- None
- [0]
- []

Yes, the answer is correct.
Score: 1
Accepted Answers:
[]

6) What is the **extra space complexity** of this approach (excluding input list)?

(/noc26_cs84/progassignment?name=628)

Week 6 Feedback Form: The Joy of Computing using Python (unit?unit=124&lesson=142)

Week 7 ()

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- (●) O(n) due to recursion stack
- ( ) O(1) because no variables are used
- ( ) O(log n)
- ( ) O(n^2)

Yes, the answer is correct.
Score: 1

Accepted Answers:
O(n) due to recursion stack

7) What is the actual role of the variable swap? ***1 point***

- ( ) It stores recursion depth
- (●) It temporarily holds a value during exchange
- ( ) It controls termination
- ( ) It accumulates reversed values

Yes, the answer is correct.
Score: 1

Accepted Answers:
It temporarily holds a value during exchange

8) For a list of length 6, how many recursive calls perform swaps? ***1 point***

- ( ) 6
- ( ) 4
- (●) 3
- ( ) 5

Yes, the answer is correct.
Score: 1

Accepted Answers:
3

9) Which statement best describes the behavior of this function? ***1 point***

- ( ) It creates and returns a new list
- ( ) It modifies the list but returns nothing
- ( ) It uses iteration internally
- (●) It mutates the original list and returns it

Yes, the answer is correct.
Score: 1

Accepted Answers:
It mutates the original list and returns it

10) If the final return helper_reverse(...) line is removed from helper_reverse, what happens? ***1 point***

- ( ) The list still gets reversed but None is returned
- ( ) The list is not reversed
- ( ) The program throws a syntax error
- (●) Only the first swap happens

Yes, the answer is correct.
Score: 1

Accepted Answers:
Only the first swap happens

Terminal-Based Tic Tac Toe Game

Background

During a programming lab session, two students decide to recreate the classic **Tic Tac Toe** game using Python. Instead of using a graphical interface or mobile application, they choose a **terminal-based approach**, relying entirely on text input and output.

The objective is to allow two players to play the game interactively while the system manages the board, validates moves, and determines the final result.

Game Environment

The game is executed in a Python environment where:

- The board consists of **nine positions arranged in three rows and three columns**
- Each position on the board can display:
  - An "X"
  - An "O"
  - Or remain empty
- The board is shown to players after every move

Players interact with the system by entering their choices through the keyboard.

Player Interaction

- The game involves **two players**
- One player uses the symbol **X**, the other uses **O**
- Players enter their moves using **row and column numbers**
- Input is provided in a fixed format during every turn
- If an invalid entry is made, the system requests a new input without progressing the game

Game Flow

- The game begins with a predefined starting player
- Players alternate turns throughout the match
- Each turn updates the visible board
- The game continues until a decisive outcome is reached
- No player is allowed to overwrite an existing move

Game Completion

The game ends in one of two ways:

1. **A winner is declared**

- o One player successfully completes a valid winning pattern
- o The game stops immediately once a winner is identified

2. **A draw is declared**
   - o All board positions are filled
   - o No player achieves a winning pattern

In both cases, the final board state is displayed before the result is announced.

The source code is presented in **three sequential parts** and must be read **in the order provided** as a single continuous program.

```
board = [[" " for _ in range(3)] for _ in range(3)]

def display_board():
    for row in board:
        print(" | ".join(row))
        if row != board[-1]:
            print("---------")

def check_won(symbol):
    for row in board:
        if all(cell == symbol for cell in row):
            return True

    for col in range(3):
        if all(row[col] == symbol for row in board):
            return True

    if all(board[i][i] == symbol for i in range(3)) or \
       all(board[i][2 - i] == symbol for i in range(3)):
        return True

    return False
```

Tic Tac Toe Game Code – Part 1

```
current_player = "X"
turns = 0

while turns < 9:
    display_board()
    move = input(f"Player {current_player}, enter your move (row and column): ").split(',')

    if len(move) != 2:
        print("Invalid input.")
        continue

    row, col = map(int, move)

    if row < 1 or row > 3 or col < 1 or col > 3:
        print("Invalid position.")
        continue

    if board[row - 1][col - 1] != " ":
        print("Cell already occupied.")
        continue

    board[row - 1][col - 1] = current_player
    turns += 1
```

Tic Tac Toe Game – Part 2

```
    if check_won(current_player):
        display_board()
        print(f"Player {current_player} wins!")
        break

    current_player = "X" if current_player == "O" else "O"
else:
    display_board()
    print("It's a draw!")
```

Tic Tac Toe Game – Part 3

11) When the program starts executing and no input has been entered yet, which of the following is true? ***1 point***

- The board is displayed with all cells filled with "X"
- The board is displayed with all cells empty
- The game immediately asks Player O for input
- The game prints “It’s a draw!”

Yes, the answer is correct.
Score: 1

Accepted Answers:
The board is displayed with all cells empty

12) A player enters the following input when prompted: ***1 point***

2

What happens next?

- The move is accepted and placed in column 2
- The program crashes with an error
- The program asks the player to re-enter input
- The turn switches to the next player

Yes, the answer is correct.
Score: 1

Accepted Answers:
The program asks the player to re-enter input

13) Which situation causes the variable turns to increase by 1? ***1 point***

- When the board is displayed
- When a player enters invalid input
- When a player overwrites an occupied cell
- When a valid move is successfully placed on the board

Yes, the answer is correct.
Score: 1

Accepted Answers:
When a valid move is successfully placed on the board

14) The game stops immediately when which of the following occurs? ***1 point***

- Nine turns have been completed
- A winning condition is detected
- An invalid move is entered
- The board is fully displayed

Yes, the answer is correct.
Score: 1

Accepted Answers:
A winning condition is detected

15) Under which condition does the program print "It's a draw!"? ***1 point***

- When Player O makes the last move
- When no diagonal matches are found
- When the loop ends without detecting a winner
- When all rows contain mixed symbols

Yes, the answer is correct.
Score: 1

Accepted Answers:
When the loop ends without detecting a winner